

A Pre-feasibility Study For Establishing A Tire Re-treading Plant In Irbid



2017



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1. Executive Summary

The project aims to establish a tire recycling plant in Irbid governorate. This project is one of the most successful industrial projects in the world. The industrial sector is one of the most important sectors in Jordan for its role in developing the economy, local market and productivity.

The basic idea of the project is to collect old used tires and use them in the manufacture of new tires with high quality and value. The use of the recycling process helps in preserving the environment and removing damaged tires that were considered burdensome and must be disposed of.

The production process is summarized in 8 stages starting with the collection and inspection of old tires and ending with the final product, which is highly efficient tires.

Table1 : Project Main Indicator

Total Investment Cost (Jordan Dinars)	5,825,888
Internal Rate of Return (IRR)	30.75%
Pay-Back Period (Years)	4
Net Present Value (Jordan Dinars)	10,286,058
Job Opportunities provided by the	58
Total Revenues (For 10 Years)	49,075,375
Total Operating Expenses (For 10 years)	26,422,783
Total Net Profit (For 10 years)	12,993,130

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

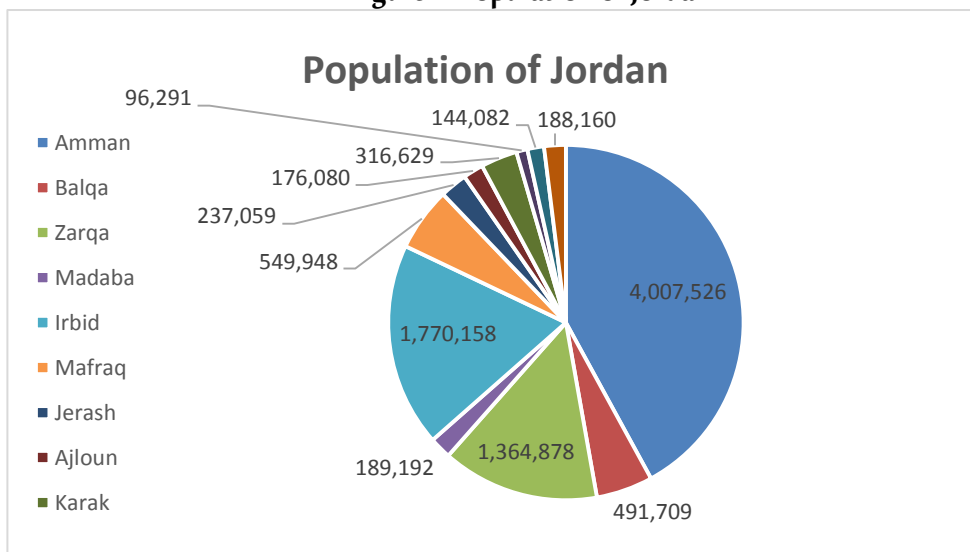
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table2 : Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

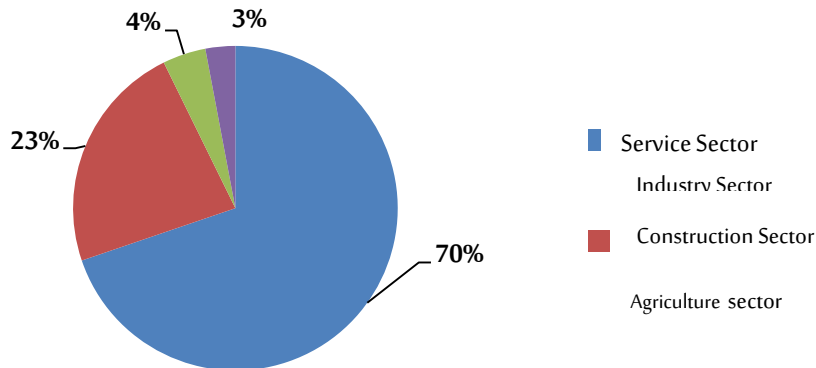
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and compromising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce. Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

3. Market Analysis

Project Description

The project aims to establish a plant for recycling and manufacturing the damaged and consumed tires and producing new high-quality tires. This project is considered a solution to a major problem, which is the accumulation of tires that are not valid for use. The process of converting old tires into rubber and then manufacturing new tires leads to achieve higher value and internal return rate.

Project Objectives

Objectives of the project are summarized as follows:

- The establishment of tire recycling plant that contributes to the preservation of the environment and reduces waste of raw materials, this method has proved its efficiency, and there are tools and equipment that have increased production quality.
- Produce new tires with high quality.
- Reduce raw materials cost
- Achieve earnings and financial returns to investors

The Proposed Product

As the planned project is intended to retread tires of heavy trucks (Lorries), 4WD, light trucks, and heavy machinery equipment (OTR), tire consumers were classified into 3 categories based on their tires needs. Consequently, market size of retreadable tires was estimated to include the following:

- Construction and contracting companies; (Heavy machinery equipment (OTR)).
- Leasing and Logistics companies; (Heavy trucks).
- Personal / Private Transportation. (Light trucks and 4WD).

Target Segments

The project targets:

- Tires shops and vehicle maintenance garages.
- Export to foreign markets.

Market Size

The following table shows the volume of exports and imports of new tires in 2015, according to the Department of Statistics:

Table 3: Imports and exports from new tires in 2015

Description	Imports (JD)	Imports (Tires)	Exports (JD)	Exports (Tires)	Jordanian Market size (Tires)	Jordanian Market size (JD)
New Tires (Light trucks and 4WD)	30,090,795	861,589	1,660,342	51,350	810,239	28,304,453
New Tires (Heavy Trucks)	2,424,739	33,325	22,334	103	33,218	2,402,405
New Tires (Machinery Equipment OTR)	1,043,417	1,884	193,683	70,241	350	849,734
Total					843,807	31,556,592

Competitors Analysis

After analyzing the market, we found that there are no any tires factories here in Jordan, whether new or recycled, therefore this project is the first of its kind in Jordan.

There are some vehicle maintenance garages and tire agencies that sell new imported tires from foreign markets.

Prices Analysis and Pricing Policy

The analysis of the market shows the suitable prices for sale are as follows, taking into account that prices will be lower than the market's prices because it will be sold to the retailers and not for customers.

- 4WD and light trucks average tire price is 45 JD.
- Heavy trucks average tire price is 150 JD.
- Heavy machinery equipment (OTR) average tire price 500 JD.

Market Share

When considering the current standards of the market, the market size of retreaded tires (recycled) mainly estimated based on the untapped market potential. Thus the size of the retreaded tires market is estimated using car market size in Jordan, which is linked directly to the number of tires. Accordingly, the estimated size of the retreaded tires market based on the following assumptions:

- Tire type: Where the market will be divided into three categories;
 - 1- Light trucks and 4WD.
 - 2- Heavy trucks.
 - 3- Heavy machinery trucks (OTR).
- The required tires:

Table4 : The number of tires in each type of trucks

The Required number of tires by each trucks type	
Type	Number of Tires
Light Trucks	5 tires (including spare tire)
4WD	5 tires (including spare tire)
Heavy Trucks	18 tires (including spare tire)
Heavy Machinery Equipment (OTR)	4 tires (including spare tire)

As stated in the Drivers and Vehicle License Department that the number of licensed cars in Jordan for 2016 is 1,429,475 cars of various types and sizes, and this number is increasing annually; this means that the market will always increase. Light trucks and 4WD comprising (74.8%) of the total cars in Jordan, Heavy trucks comprising (21%) and heavy machinery equipment has the lowest percent by (4.2%). Based on the previous information, The Jordanian market size is about 6 million tires from the different types. And the total production capacity of the project was estimated to 80,000 tires / year from all types.

Projected Revenues Analysis

Based on the Jordanian market, which was 843,807 Tires in 2015 and the production capacity of the plant which is 80,000 Tires from all types and sizes. The following table shows the projected revenues for 10 years from the starting date of the project:

Table 5: Projected Revenues (2018-2027)

Prices and Revenues	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
4WD and Light Trucks Average Tire Prices	45.00	46.35	47.74	49.17	50.65	52.17	53.73	55.34	57.00	58.71
Heavy Trucks Average Tire Prices	150.00	154.50	159.14	163.91	168.83	173.89	179.11	184.48	190.02	195.72
OTR Average Tire Prices	500.00	515.00	530.45	546.36	562.75	579.64	597.03	614.94	633.39	652.39
Sell Proportion of Full Production Capacity	40%	45%	50%	55%	60%	60%	70%	70%	80%	80%
4WD and Light Trucks Tires Revenues	1077120	1248113	1428396	1618372	1818462	1873016	2250741	2318263	2728927	2810795
Heavy Trucks Tires Revenues	1008000	1168020	1336734	1514520	1701769	1752822	2106308	2169497	2553808	2630423
OTR Tires Revenues	672000	778680	891156	1009680	1134513	1168548	1404206	1446332	1702539	1753615
Revenues	2757120	3194813	3656286	4142572	4654744	4794387	5761255	5934092	6985274	7194832

4. Techniacal Analysis

The Required Human Resources

Table 6 : Required Labor

Job Title	Number
Indirect Employees	
Plant Manager	1
Financial Managel	1
Accountant	4
Marketing Officer	1
Marketing Employees	4
HR Officer	1
HR Employees	2
Cleaners	4
Total	18
Direct Employees	
Supervisors	3
Warehouses Officer	1
Warehouses Employees	3
Maintenance Technician	1
Driver	2
Production Labor	30
Total	58

The following is a clarification of the functions of the workforce:

- **Financial and Administrative Affairs**
 - Ensure the proper implementation of financial policies and procedures.
 - Record all financial transactions and confirm the correct implementation of procedures.
 - Record and control registers relating to cash management (cash on hand, cash at banks, credit facilities and bank guarantees). Do studies as well to reduce costs.

- Organize and archive all financial documents and constantly updated in an orderly manner.
- Follow-up on all human resources procedures consistently and recruitment, training and salaries, vacations and other.
- Follow-up on the readiness of information technology.
- **Purchasing and Warehouses**
 - Check the actual inventory of raw materials and spare parts and make Observations about the levels of differences between the accounting numbers and actual quantities.
 - Carry out the purchase and supply in the factory, including a review of procurement and supplier prices, wages and shipping processes requests for delivery, distribution of procurement functions, provide advice in order to ensure alignment of materials and services to the approved specifications.
 - Analyze supply requirements, developing resources, recommending the process of analyzing and evaluating suppliers and keep a list of approved suppliers in line with the company's adopted procedures and policies
 - Ensure the preparation of Procurement status reports, provide a summary of progress and submit reports on the prices of materials and industrial trends
- **Production**
 - Control the activities of material and production staff, maintenance, safety and quality
 - Examine the project site and ensure adherence to the design specifications and standards of public safety.
 - Review and analyze the production plans, preventive maintenance plans, material request orders and reports of quality control and testing.
 - Plan and implement the standard operating procedures, developing the key performance indicators and monitoring these indicators.
 - Ensure and control the production processes and provide the required human resources. Ensure that key elements are constantly trained and developed in order to increase productivity.

- Follow-up on functions of various units that affect the production and ensure problem-solving between different functions which guarantee the smooth distribution of tasks and be in line with the needs of sales and marketing and factory goals .
- **Maintenance**
 - Perform all the maintenance operations at the plant by the maintenance staff
 - The development of preventive and corrective maintenance plans for production lines and ensure compliance with these plans
 - Coordinate with the production and quality control departments regarding the timing and type of maintenance without affecting the target production volume
 - Coordinate with maintenance services providers to request for spare parts or doing maintenance for machines under guarantee

Site Analysis

The proposed site for this project is Al Hasan Industrial City in Irbid, which is the industrial investment center in the north.

Technical Requirements Analysis

Land and construction works

The project will be constructed on land with an area of 12,000 sqm. Divided into the factory, warehouses, parking spaces and administration offices

The following table shows the required land:

Table7 : The required land

Land	Area (sqm)
The Required land	12,000

The required construction works for the project is represented as follows:

Construction Works	Area (sqm)
Hangar	5,000
Designing and drawing	5,000
Paving the ground	1,000

Machinery and Equipment

The following table provides the machinery and equipment needed for the project:

Machinery and Equipment	Quantity
Pressure testing Machine for initial and final inspection	1
Inspection spreader machine for truck tires	1
Buffing machine for light truck, 4x4 and truck tires	1
Dust absorption system for above buffing machine	2
Tire repair system	1
Complete set of tools for repairs, skive filling and cementing of truck tires	1
Extruder builder for truck tires	2
Jib crane for truck tires	1
Table for application of sidewall veneer	1
bladder type steam heated presses suitable to retread tires	1
Tire lifters (one each two machines)	2
Pressing Machine	1
Cabin for painting the finished tires	1
Table with adjustable height for repairs	1
Big Sizes Tires Cutting and Controlling Machine	1
Big Size Dust Absorption Machine	1
Big Sizes Tires Supporting Machine	1
Big Sizes Tires Repairing and Fixing Tools	1

Machinery and Equipment	Quantity
Air Dragging Machine	1
Exhaust covers with fan	1
Extruder Builder for O.T.R Tires	1
Thermal Compression Machine (Small)	1
Thermal Compression Machine (Meduim)	1
Thermal Compression Machine (Large)	1
Jib crane lifters for OTR Presses	1
O.T.R Tires Pressing Machine	1
Set of aluminium bead plates	1
Special Sizes Tires Pressing Machine	1
Autoclave Chamber	1
Secondary Pressure Measuring Machine	1
Big Sizes Tires Cutting and Controlling Machine	1
Big Sizes Tires Grinding and Polishing Machine	1
Rapid dissolution machine composed by column	1
Stitching Machine	1
Water Steam Tank	1
Hydraulic Pressing Machine	1

Vehicles and Transportation

The project needs Vehicles as following:

Table8 : Vehicles and Transportation

Vehicles	Quantity
Pickup Trucks	2

Raw Materials Requirements

Table 9 : The Required Raw Materials for tires retreading

Raw Materials	Avg. Price (JD/Kg)
Rubber Materials	1.35

Tire-retreading Process

Tires-retreading process in the proposed project will consist of eight different phases according to the following:

Tire-retreading Process

Tire-retreading Project

1. **INSPECTION:** This is the first phase of the process. It consists of the selection of the collected casings (used tires) and in the determination of their destination: suitable for retreading process or to be destined for different disposal.
2. **BUFFING:** The casings selected as suitable for the retreading process are sent to the buffing department for removal of the residual of old tread pattern.
3. **REPAIRING:** The buffed casings are sent to the repair department for various operations, such as grinding, cleaning, application of repair patches, etc.
4. **CEMENTING AND SKIVE FILLING:** This operation consists in the application of the cement (glue) and the filling of the holes generated in the previous repair phase. The holes are filled with small hand held gun extruders.
5. **AUTOCLAVE CURING (O.T.R. REPAIRS ONLY):** The OTR tires that are not designated to the retreading process, but do only have to be repaired and then put back in production, are sent to this department after application of repairs. In case of OTR tires are often damaged but that they still have a lot of working hours in their tread. Therefore, instead of sending them for retreading, i.e. replacement of the whole tread, they are sent for repair only. The repair patch applied to the damaged area needs to be cured and this is usually done inside an autoclave chamber.
6. **EXTRUDER BUILDER AND SIDEWALL VENEERING:** The cemented and repaired casings are sent to the building department for application of the new rubber coat. This operation is done with an automatic extruder builder, usually CNC controlled, that will apply a suitable shape and thickness of rubber to the casing.

7. **CURING (VULCANIZING):** The curing of the so prepared tires is made inside curing pressers (hot cure process). During this phase the new tread pattern is stamped onto the built casing. We have suggested the hot cure process in our initial quotation, but there is also the possibility to integrate the curing process with autoclaves allowing vulcanizing the tires with the pre-cure system. The decision weather to make this integration or not, depends on various factors of the local market and will be evaluated after receiving your replies to our questionnaire. For the moment, we have only foreseen to have curing pressers for hot cure process.
8. **PAINTING AND FINISHING:** Once the tires have been cured, they are sent to the final inspection department, where they can also be painted and cleared from rubber flashes and other visual minor defects that are eliminated before sending the tires to the finished product storage area.

5. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 12.05% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 10: General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 11: Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 12: Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table13 : Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	35.00%
Utilities Expenses from Total Revenues	5.0%
Maintenance Expenses from Total Revenues	2.50%
Depreciation Expenses from Total Revenues	1.00%
Insurance Expenses from Total Assets Value	1.00%
Marketing Expenses from Total Revenues	0.25%

Table 14: Income Tax Assumptions

Income Tax Assumptions	Value
Average Income Tax in Jordan(1)	5%
Income Tax Deduction(2)	0%
Income Tax after Deduction	5%
Compulsary Reserve Percentage(3)	10%
Other Assumptions	Value
Annual Monthly Salaries have been calculated after multiplying monthly salaries	16

Table 15: Risk Premuim

Risk Premuim	
Risk Free Rate of Return (1)	6.50%
Return to Debt Maturity (2)	9.48%
Market Risk Premuim (3)	10.12%
Income Tax Rate	5.00%
Beta (4)	1
Growth rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 16: Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premuim	%10.12
Income Tax Rate	%5
Bets	1.00
Equity	2,330,355
Loans Value	3,495,533
Loans	
Cost of Borrowing before Tax	%9.48

Cost of Borrowing After Tax	%9.01
Loans Value	3,495,533
Loans Percentage	%60
Equity	
Cost of Equity	%16.6
Equity Value	2,330,355
Equity Percentage	%40
Gross	
Project Value	5,825,888
Proportion of the project	%100

Capital Expenditures

The estimated cost of the project is 5,825,888 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 17: Land Capital Expenditure

Land	Cost) Jordan Dinar/Meters Square(	Area (M2)	Cost
Required Land	5	12,000	60,000
Total			60,000

Table 18: Construction Work Capital Expenditure

Description	Cost (JOD per Squared	Area	Gross
Hangar	100	5,000	500,000
Construction Works	150	5,000	750,000
Paving	17	1,000	17,000
Others	-	-	126,700
Gross Total			1,393,700

Table 19: Machinery and Equipment Capital Expenditure

Machinery and equipment	Quantity	Cost / JD	Total Cost
Pressure testing Machine for initial and final inspection	1	42,000	42,000
Inspection spreader machine for truck tires	1	6,600	6,600
Buffing machine for light truck, 4x4 and truck tires	1	48,000	48,000
Dust absorption system for above buffing machine	2	7,600	15,200
Tire repair system	1	21,500	21,500
Complete set of tools for repairs, skive filling and cementing of truck tires	1	35,000	35,000
Extruder builder for truck tires	2	83,000	166,000
Jib crane for truck tires	1	18,000	18,000
Table for application of sidewall veneer	1	12,500	12,500
bladder type steam heated presses suitable to retread tires	1	470,000	470,000
Tire lifters (one each two machines)	2	9,300	18,600
Pressing Machine	1	155,000	155,000
Cabin for painting the finished tires	1	47,000	47,000
Table with adjustable height for repairs	1	7,800	7,800
Big Sizes Tires Cutting and Controlling Machine	1	30,000	30,000
Big Size Dust Absorption Machine	1	137,500	137,500
Big Sizes Tires Supporting Machine	1	75,000	75,000
Big Sizes Tires Repairing and Fixing Tools	1	40,500	40,500
Air Dragging Machine	1	15,000	15,000
Exhaust covers with fan	1	28,500	28,500
Extruder Builder for O.T.R Tires	1	290,000	290,000
Thermal Compression Machine (Small)	1	165,000	165,000
Thermal Compression Machine (Meduim)	1	181,000	181,000
Thermal Compression Machine (Large)	1	245,100	245,100
Jib crane lifters for OTR Presses	1	72,000	72,000
O.T.R Tires Pressing Machine	1	101,365	101,365
Set of aluminium bead plates	1	10,000	10,000

Machinery and equipment	Quantity	Cost / JD	Total Cost
Special Sizes Tires Pressing Machine	1	198,000	198,000
Autoclave Chamber	1	235,000	235,000
Secondary Pressure Measuring Machine	1	15,000	15,000
Big Sizes Tires Cutting and Controlling Machine	1	13,500	13,500
Big Sizes Tires Grinding and Polishing Machine	1	7,600	7,600
Rapid dissolution machine composed by column	1	13,200	13,200
Stitching Machine	1	12,500	12,500
Water Steam Tank	1	190,500	190,500
Hydraulic Pressing Machine	1	55,000	55,000
Gross Total			3,194,465

Table 20: Furniture Capital Expenditure

Furniture and furnishings	Quantity	Cost / JD	Total Cost
Furniture	-	-	20,000
Total			20,000

Table 21: Transporation and Vehicles Capital Expenditure

Transporation and Vehicles	Quantity	Unit Cost/JOD	Total Cost/JOD
Pickup Truck	2	23,000	46,000
Gross Total			46,000

Table 22: Pre Operating Expenses

Pre-operating expenses	Estimaited cost (JD)
Transportation	1,500
Marketing	1,500
Training	1,500
Total	4,500

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 23: Initial Working Capital

Working Capital	No. of	%	Estimated Cost
Operating expenses	3	%25	390,760
General and administrative expenses	3	%25	16,713
Marketing expenses	3	%25	1,723
Indirect Salaries	3	%25	48,400
Total			457,597

Table 24: Expenses Summary

Capital expenditure	Cost (JD)	Contingency	Cost (JD)	%
Land	180,000	10%	198,000	%3.4
Construction works	1,393,700	10%	1,533,070	%26.3
Machinery & Equipment	3,194,465	10%	3,513,912	%60.3
Furniture	20,000	10%	22,000	%0.4
Vehicles	46,000	10%	50,600	%0.9
Pre-Operating Expenses	4,500	10%	4,950	%0.1
Working capital	457,597	10%	503,357	%8.6
Total (JD)	5,296,262		5,825,888	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (5,825,888) JD, where 60% of the project will be financed through loans (3,495,533) JD, and the remaining (40%) through (Equity) with (2,330,355) JD. The following table summarizes the general structure of the required financing:

Table 25: Sources of Funding

Funding Sources	Amount	Percentage
Equity (Self Financing)	2,330,355	%40
Loans	3,495,533	%60
Total	5,825,888	%100
Use of Fund	Value	Ratio
Capital Expenditures	5,317,582	%91.28
Pre-Operating Expenses	4,950	%0.08
Working Capital	503,357	%8.64
Total	5,825,888	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 26: Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging from	964,992	1,118,18	1,279,70	1,449,90	1,629,16	1,678,03	2,016,43	2,076,93	2,444,84	2,518,19
Utilities from Revenues	137,856	159,741	182,814	207,129	232,737	239,719	288,063	296,705	349,264	359,742
Maintenance from Revenues	68,928	79,870	91,407	103,564	116,369	119,860	144,031	148,352	174,632	179,871
Disposables from Revenues	27,571	31,948	36,563	41,426	46,547	47,944	57,613	59,341	69,853	71,948
Direct Salaries	221,600	230,686	240,144	249,990	260,239	270,909	282,016	293,579	305,616	318,146
Others	142,095	162,043	183,063	205,201	228,505	235,647	278,816	287,491	334,421	344,790
Total	1,563,04	1,782,47	2,013,69	2,257,20	2,513,55	2,592,11	3,066,97	3,162,40	3,678,63	3,792,68

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 27: General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	2,400	2,472	2,546	2,623	2,701	2,782	2,866	2,952	3,040	3,131
Hospitality	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Stationary and Printing	600	618	637	656	675	696	716	738	760	783
Transportation	2,400	2,472	2,546	2,623	2,701	2,782	2,866	2,952	3,040	3,131
Training	1,000	1,030	1,061	1,093	1,126	1,159	1,194	1,230	1,267	1,305
Insurance	53,176	54,771	56,414	58,107	59,850	61,645	63,495	65,400	67,362	69,382
Miscellaneous	6,078	6,260	6,448	6,641	6,840	7,046	7,257	7,475	7,699	7,930
Total	66,853	68,859	70,925	73,053	75,244	77,501	79,826	82,221	84,688	87,229

Marketing Expenses

Table 28: Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	6,893	7,987	9,141	10,356	11,637	11,986	14,403	14,835	17,463	17,987

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 29: Expected monthly salaries (2018-2027)

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
General Manager	2,500	2,603	2,709	2,820	2,936	3,056	3,182	3,312	3,448	3,589
Financial Manager	1,750	1,822	1,896	1,974	2,055	2,139	2,227	2,318	2,413	2,512
Accountant	500	521	542	564	587	611	636	662	690	718
Marketing Manager	1,250	1,301	1,355	1,410	1,468	1,528	1,591	1,656	1,724	1,795
Marketing Employees	450	468	488	508	528	550	573	596	621	646
Human Resources Manager	1,000	1,041	1,084	1,128	1,174	1,223	1,273	1,325	1,379	1,436
Human Resources Employees	400	416	433	451	470	489	509	530	552	574
Cleaner	250	260	271	282	294	306	318	331	345	359
Direct Employees										
Supervisor	750	781	813	846	881	917	954	994	1,034	1,077
Warehouses Manager	1,200	1,249	1,300	1,354	1,409	1,467	1,527	1,590	1,655	1,723
Warehouses Employees	300	312	325	338	352	367	382	397	414	431
Maintenance Employees	500	521	542	564	587	611	636	662	690	718

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Driver	300	312	325	338	352	367	382	397	414	431
Worker	280	291	303	316	329	342	356	371	386	402

Table 30: Expected yearly salaries for the years (2018-2027)

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
General Manager	40,000	41,640	43,347	45,124	46,975	48,901	50,905	52,993	55,165	57,427
Financial Manager	28,000	29,148	30,343	31,587	32,882	34,230	35,634	37,095	38,616	40,199
Accountant	32,000	33,312	34,678	36,100	37,580	39,120	40,724	42,394	44,132	45,942
Marketing Manager	20,000	20,820	21,674	22,562	23,487	24,450	25,453	26,496	27,583	28,714
Marketing Employees	28,800	29,981	31,210	32,490	33,822	35,208	36,652	38,155	39,719	41,347
Human Resources Manager	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
Human Resources Employees	12,800	13,325	13,871	14,440	15,032	15,648	16,290	16,958	17,653	18,377
Cleaner	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
Total Indirect Salaries	193,600	201,538	209,801	218,402	227,357	236,679	246,382	256,484	267,000	277,947
Direct Employees										
Supervisor	36,000	37,476	39,013	40,612	42,277	44,010	45,815	47,693	49,649	51,684
Warehouses Manager	19,200	19,987	20,807	21,660	22,548	23,472	24,435	25,436	26,479	27,565

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Warehouses Employees	14,400	14,990	15,605	16,245	16,911	17,604	18,326	19,077	19,860	20,674
Maintenance Employees	8,000	8,328	8,669	9,025	9,395	9,780	10,181	10,599	11,033	11,485
Driver	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Worker	134,400	139,910	145,647	151,618	157,835	164,306	171,042	178,055	185,355	192,955
Total Direct Salaries	221,600	230,686	240,144	249,990	260,239	270,909	282,016	293,579	305,616	318,146
Total Salaries	415,200	432,223	449,944	468,392	487,596	507,588	528,399	550,063	572,616	596,093

Depreciations

:The following tables show all equipment costs, expenses and depreciation rates

Table 31: Capex and Depreciation Expenses

Costs of equipment, capital expenditures and annual depreciation	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	198,000	%0.0	%0.0
Construction Works	1,533,070	%5.0	%0.0
Machinaries	3,513,912	%10.0	%2.0
Furniture	22,000	%10.0	%5.0
Vehicles	50,600	%10.0	%0.0
	5,317,582		

Table 32: Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciations	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	5,317,58	5,388,96	5,461,79	5,536,12	5,611,98	5,689,39	5,768,38	5,849,00	5,931,28	6,015,24
Total Depreciation Expenses	435,305	442,442	449,726	457,159	464,745	472,486	480,385	488,447	496,675	505,071
Total Accumulated Depreciation	435,305	877,747	1,327,47	1,784,63	2,249,37	2,721,86	3,202,24	3,690,69	4,187,37	4,692,44
Total Additions	0	71,378	72,839	74,330	75,853	77,408	78,997	80,619	82,275	83,967
Total Net Book Values	4,882,27	4,511,21	4,134,32	3,751,49	3,362,60	2,967,52	2,566,13	2,158,31	1,743,91	1,322,80

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 33: Loan Details

Loan Value	3,495,533			
Annual Interest Rate	9.48%			
Loans Period	5			
Loan Starts at year:	2018			
Annual Payment	909,895			
Number of Payments	5			

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				3,495,533
2019	909,895	331,377	578,519	2,917,014
2020	909,895	276,533	633,362	2,283,652
2021	909,895	216,490	693,405	1,590,246
2022	909,895	150,755	759,140	831,106
2023	909,895	78,789	831,106	0

Income Statement

Table 34: Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	2,757,12	3,194,81	3,656,28	4,142,57	4,654,74	4,794,38	5,761,25	5,934,09	6,985,27	7,194,832
Gross Operating Revenues	2,757,12	3,194,81	3,656,28	4,142,57	4,654,74	4,794,38	5,761,25	5,934,09	6,985,27	7,194,832
Operating Expenses	1,563,04	1,782,47	2,013,69	2,257,20	2,513,55	2,592,11	3,066,97	3,162,40	3,678,63	-3,792,688
Gross Operating Profit	1,194,07	1,412,34	1,642,59	1,885,36	2,141,18	2,202,27	2,694,27	2,771,69	3,306,64	3,402,145
Gross Profit Percentage	%43	%44	%45	%46	%46	%46	%47	%47	%47	%47
Salaries and Benefits (Indirect	-193,600	-201,538	-209,801	-218,402	-227,357	-236,679	-246,382	-256,484	-267,000	-277,947
General and Administraive	-66,853	-68,859	-70,925	-73,053	-75,244	-77,501	-79,826	-82,221	-84,688	-87,229
Markeing Expenses	-6,893	-7,987	-9,141	-10,356	-11,637	-11,986	-14,403	-14,835	-17,463	-17,987
Pre Operating Expenses	-4,950									
Gross Indirect Expenses	-272,296	-278,384	-289,866	-301,811	-314,238	-326,166	-340,612	-353,541	-369,151	-383,163
Income before Interest, Depreciation	921,782	1,133,95	1,352,72	1,583,55	1,826,94	1,876,10	2,353,66	2,418,15	2,937,49	3,018,982
Fixed Assets Depreciations	-435,305	-442,442	-449,726	-457,159	-464,745	-472,486	-480,385	-488,447	-496,675	-505,071
Income before Tax and Interests	486,477	691,515	903,002	1,126,39	1,362,20	1,403,62	1,873,27	1,929,70	2,440,81	2,513,911
Bank Interests	-331,377	-276,533	-216,490	-150,755	-78,789	-0	-0	-0	-0	-0

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Income Before Tax	155,101	414,982	686,512	975,636	1,283,41	1,403,62	1,873,27	1,929,70	2,440,81	2,513,911
Income Tax	-7,755	-20,749	-34,326	-48,782	-64,171	-70,181	-93,664	-96,485	-122,041	-125,696
Net Profit	147,346	394,233	652,187	926,855	1,219,24	1,333,44	1,779,61	1,833,21	2,318,77	2,388,215
Net Profit Percentage	%5	%12	%18	%22	%26	%28	%31	%31	%33	%33
Compulsory Reserves	-14,735	-39,423	-65,219	-92,685	-121,924	-133,344	-177,962	-183,322	-231,878	-238,822
Retained Earnings	132,611	487,420	1,074,38	1,908,55	3,005,87	4,205,97	5,807,62	7,457,52	9,544,42	11,693,81

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 35: Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash flows from operating										
Net profit (loss)	147,346	394,233	652,187	926,855	1,219,244	1,333,440	1,779,615	1,833,219	2,318,777	2,388,215
Non-monetary adjustments										
Bank benefits	331,377	276,533	216,490	150,755	78,789	0	0	0	0	0
Consumption	435,305	442,442	449,726	457,159	464,745	472,486	480,385	488,447	496,675	505,071
Operating cash flows before changes in working capital	914,027	1,113,208	1,318,403	1,534,769	1,762,777	1,805,926	2,260,000	2,321,666	2,815,451	2,893,287
Shortage (increase) in inventory	-80,416	-12,766	-13,460	-14,183	-14,938	-4,073	-28,200	-5,041	-30,659	-6,112
Decrease (increase) in accounts	-229,760	-36,474	-38,456	-40,524	-42,681	-11,637	-80,572	-14,403	-87,599	-17,463
(Decrease) increase in accounts	99,675	11,040	11,617	12,220	12,848	4,947	22,743	5,821	24,720	6,785
Change in working capital	-210,501	-38,201	-40,299	-42,487	-44,771	-10,763	-86,029	-13,623	-93,538	-16,790
Net cash flows (used) from operating activities	703,526	1,075,007	1,278,105	1,492,282	1,718,006	1,795,163	2,173,971	2,308,044	2,721,913	2,876,497
Cash flows from investing										
(Purchase) of fixed assets	-5,317,582	-71,378	-72,839	-74,330	-75,853	-77,408	-78,997	-80,619	-82,275	-83,967

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net flows (used) from investment activities	-5,317,582	-71,378	-72,839	-74,330	-75,853	-77,408	-78,997	-80,619	-82,275	-83,967
Cash flows from financing activities										
Equity capital	2,330,355									
Loan Amortization	-578,519	-633,362	-693,405	-759,140	-831,106	0	0	0	0	0
Bank Interest Rate	-331,377	-276,533	-216,490	-150,755	-78,789	-0	-0	-0	-0	-0
Loans	3,495,533	0	0	0	0	0	0	0	0	0
Net cash flows (used) from financing activities	4,915,993	-909,895	-909,895	-909,895	-909,895	0	0	0	0	0
Net increase (decrease) in cash	301,937	93,734	295,371	508,056	732,258	1,717,754	2,094,974	2,227,425	2,639,638	2,792,529
Cashflows at the Beginning of	0	301,937	395,671	691,041	1,199,098	1,931,355	3,649,109	5,744,084	7,971,509	10,611,146
Cashflows at the End of Period	301,937	395,671	691,041	1,199,098	1,931,355	3,649,109	5,744,084	7,971,509	10,611,146	13,403,675

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 36: Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current assets										
Fund	301,937	395,671	691,041	1,199,098	1,931,355	3,649,109	5,744,084	7,971,509	10,611,146	13,403,675
Inventory	80,416	93,182	106,642	120,825	135,763	139,836	168,037	173,078	203,737	209,849
accounts receivable	229,760	266,234	304,690	345,214	387,895	399,532	480,105	494,508	582,106	599,569
Total assets traded	612,113	755,087	1,102,373	1,665,137	2,455,014	4,188,478	6,392,225	8,639,094	11,396,990	14,213,094
Non-current assets										
Fixed assets (net)	4,882,277	4,511,213	4,134,325	3,751,496	3,362,604	2,967,527	2,566,139	2,158,311	1,743,911	1,322,807
Total non-current assets	4,882,277	4,511,213	4,134,325	3,751,496	3,362,604	2,967,527	2,566,139	2,158,311	1,743,911	1,322,807
Total assets	5,494,390	5,266,300	5,236,699	5,416,633	5,817,618	7,156,005	8,958,364	10,797,405	13,140,901	15,535,901
Current liabilities										
Payables	99,675	110,715	122,332	134,552	147,400	152,346	175,090	180,911	205,631	212,416
Remaining amount of Loan	633,362	693,405	759,140	831,106	0-	0-	0-	0-	0-	0
Total current liabilities	733,037	804,120	881,472	965,658	147,400	152,346	175,090	180,911	205,631	212,416

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Non-current liabilities										
long term loans	2,283,652	1,590,246	831,106	0	0	0	0	0	0	0
Total long - term liabilities	2,283,652	1,590,246	831,106	0	0	0	0	0	0	0
Total liabilities	3,016,689	2,394,366	1,712,578	965,658	147,400	152,346	175,090	180,911	205,631	212,416
Property rights										
Members Contributions	2,330,355	2,330,355	2,330,355	2,330,355	2,330,355	2,330,355	2,330,355	2,330,355	2,330,355	2,330,355
Compulsory reserve	14,735	54,158	119,376	212,062	333,986	467,330	645,292	828,614	1,060,491	1,299,313
Profit (loss)	132,611	487,420	1,074,388	1,908,558	3,005,877	4,205,973	5,807,627	7,457,524	9,544,423	11,693,817
Total equity	2,477,701	2,871,934	3,524,120	4,450,975	5,670,219	7,003,659	8,783,274	10,616,493	12,935,270	15,323,485
Total liabilities and equity	5,494,390	5,266,300	5,236,699	5,416,633	5,817,618	7,156,005	8,958,364	10,797,405	13,140,901	15,535,901

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 37: Free Net Cash flows Table

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	القيمة المهائية
Net Free Cashflows	-5,825,888	708,476	1,003,62	1,205,26	1,417,95	1,642,15	1,717,754	2,094,97	2,227,42	2,639,63	2,792,52	36,070,22
Discount Factor	1.00	0.71	0.63	0.57	0.51	0.45	0.40	0.36	0.32	0.29	0.26	0.26
Net Present Value for Cash Flows	-5,825,888	503,583	636,650	682,327	716,396	740,435	691,220	752,343	713,875	754,998	712,822	9,207,296

Table below (37) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 38: payback period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	-5,825,888	708,476	1,003,629	1,205,266	1,417,952	1,642,153	1,717,754	2,094,974	2,227,425	2,639,638	2,792,529
Project Value	5,825,888	5,117,412	4,113,783	2,908,518	1,490,566	-151,587	1,869,341	3,964,316	6,191,741	8,831,378	11,623,907
Payback Period	4	1	1	1	1	0	0	0	0	0	0

Table 39: Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%12.05
Net Present Value for Cashflows	10,286,058
Payback Period	4
Internal Rate of Return	%30.75

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 10,286,058 Jordan Dinars considering that the project provides 41 Job Opportunities for the governorate residents.

Table 40: Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%30.75	4	12.05%
Revenues declined by 10%	%28.08	5	12.05%
Operating Expenses Increased by 10%	%27.74	5	12.05%